

Pneumonia detection from chest X-rays using convolutional neural networks

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Abstract. This project investigates the use of convolutional neural networks (CNNs) for classifying chest X-ray images as either "normal" or "pneumonia". A publicly available Kaggle dataset of clinical X-rays[4] was preprocessed, normalised and resized to ensure consistency before training. A simplified CNN model was implemented to learn visual features directly from pixel data and compared against a logistic regression baseline classifier. The CNN achieved a higher recall and F1-score, demonstrating the value of convolutional feature extraction for medical image analysis[1][5]. However, the model produced a large number of false positives and showed signs of overfitting, suggesting limited generalisation[3]. These results highlight both the potential and limitations of using a basic CNN architecture for pneumonia detection, and suggest that larger datasets or stronger augmentation strategies could improve reliability in future work.

1 Introduction

Pneumonia is a respiratory infection that causes inflammation of the air sacs within the lungs. The most common ways of diagnosing pneumonia are chest X-rays, blood tests, or pulse oximetry, with chest X-rays being the primary tool used to support a clinical diagnosis. Interpretation of these radiographs can vary as it relies on clinician expertise and can be affected by workload, image quality or subtle visual patterns that are not always easy to identify. A computational aid can reduce this variability by providing consistent and repeatable assessments.

Deep learning has become an active area of research in medical imaging [3], with multiple studies demonstrating that convolutional neural networks (CNNs) can identify abnormalities in chest X-rays at performance levels comparable to experienced clinicians[1][8]. CNNs are capable of learning patterns found from pixel data, making them suitable for distinguishing between healthy lungs and lungs that show signs of infection. This task can be expressed as the following binary classification function

$$f_{\theta}(X) \rightarrow \{\text{normal}, \text{pneumonia}\}, \quad (1)$$

where X is an input chest X-ray image and f_{θ} is the trained model function.

Training models on a large-scale dataset allows fast and reproducible assessments that complements existing clinical workflows. As a result, CNN-based systems are beginning to be used as a supportive tool that can provide a second opinion. This project follows this direction by implementing a simple CNN model on a pub-

licly available pneumonia dataset[4] and evaluating its performance against a basic baseline classifier.

2 Background

Deep learning has become widely recognised within medical image analysis due to its ability to extract key features from pixel data without the need to define descriptors. Traditional image processing methods rely on manually designed descriptors such as edge detectors. These descriptors often failed to capture the subtle differences in medical scans[7]. However, convolutional neural networks (CNNs) learn these patterns directly during the training phase, making them well suited for classification tasks.

A CNN extracts information using convolutional filters, where each filter slides across the image to detect specific local patterns. Although CNN layers are conventionally described as performing convolution, most deep learning frameworks actually implement cross-correlation[2] [6].

The cross-correlation equation can be denoted by:

$$S(i, j) = (I * K)(i, j) = \sum_m \sum_n I(i + m, j + n)K(m, n) \quad (2)$$

While the convolutional equation is denoted by:

$$S(i, j) = (I * K)(i, j) = \sum_m \sum_n I(i - m, j - n)K(m, n) \quad (3)$$

Cross-correlation is preferred in modern deep learning frameworks because it is easier to compute and avoids the additional kernel flipping step required by convolution. Despite the differences, cross-correlation is still able to measure how strongly a learned filter responds to patterns in specific regions of the input image. The output of this process $S(i, j)$ is a feature map where higher values indicate that a particular feature has been detected.

CNNs are able to build on these feature maps by stacking multiple layers. Early layers will normally detect simple structures such as edges or corners, while the deeper layers can broaden out and are able to detect shapes and textures which are found in diseased lungs. The ability to layer allows the model to learn useful representations at different levels of detail, which is especially important for medical images where visual differences can be subtle.

Prior research has shown that these learned representations make CNNs highly effective for pneumonia detection. Early work demonstrated strong performance on large public datasets of chest X-rays.

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More recent studies build on this idea by using deeper architectures such as ResNet or DenseNet to increase accuracy and robustness. These ideas form the foundation for the simplified CNN model implemented in this project, which is designed to remain computationally efficient while capturing the essential behaviour of deeper architectures.

3 Experiments and results

The dataset used in this project is publicly available and contains a collection of real clinical X-ray images labelled as either "normal" or "pneumonia"[4]. The images are provided in varying resolutions, so all samples were resized to 224×224 to make the dataset consistent before training.

The dataset follows a standard structure with separate folders for training, validation and testing. This division prevents data leakage and ensures that the model is evaluated fairly on unseen images.

Before training the model, all images have to be preprocessed to ensure they are consistent and suitable for the CNN input format. Each image is then converted to a PyTorch tensor[9], which allows pixel values to be represented numerically for model input. The pixel values were then normalised to a standard range to help stabilise the training process and ensure that features across images were treated consistently.

Once the images had been resized, simple transformations such as random horizontal flips were performed to help the model better generalise by exposing it to variations of the same image. The primary purpose for this was to reduce the risk of overfitting, which is the result of the model memorising training data instead of learning to generalise underlying patterns that may be present. No heavy augmentations were used, as the goal was to keep the experimental setup straightforward and aligned with the simplified model being implemented.

The model used in this project is a simple convolutional neural network designed to classify chest X-ray images into either "normal" or "pneumonia". The architecture follows a standard CNN structure with multiple convolutional layers, each followed by a ReLU activation function and a max pooling layer. These layers allow the network to gradually reduce the spatial size of the feature maps while increasing the depth, giving it the ability to learn useful patterns from the images.

ReLU (Rectified Linear Unit) is used as the activation function because it introduces non-linearity into the network by setting negative values to zero. This makes it easier for the model to learn complex patterns without adding extra computational cost. Max pooling is used between convolutional layers and works by selecting the maximum value within small regions of the feature maps. This reduces their resolution, making the model more robust to spatial variations, such as slight shifts or noise in the input image.

As data moves through the convolutional layers, the network will first learn simple features such as edges or corners, and as it progresses through the layers it will eventually be able to learn more complex shapes and textures that may be associated with infected lung regions. After the final convolutional block, the resulting feature maps are flattened and passed into fully connected layers, which combine the extracted features to produce a final prediction. The last layer outputs two values which represent either "normal" or "pneumonia", and a softmax function is applied to convert these values into probabilities that sum to one.

A simplified architecture was chosen for this project to understand and demonstrate the core behaviour of a CNN instead of trying to

build the most optimised solution.

The model was trained using supervised learning, where each X-ray image is paired with a label indicating whether it is "normal" or "pneumonia". The loss function used for training was cross-entropy loss, which is standard for multi-class classification tasks and measures how well predicted probabilities match the true labels.

The network parameters were optimised using the Adam optimiser, as it provides an adaptive learning rate and performs well in practice for deep learning tasks. The learning rate was set to 0.001, which offered a good balance between stability and convergence speed during training. The model was trained for five epochs, where one epoch represents a complete pass through the entire training dataset. During training, the validation accuracy was tracked to monitor progress and detect when performance stopped increasing.

The batch size was set to 32 during training to ensure efficient use of memory while still providing stable gradient estimates. To evaluate the performance, the training and validation accuracy were recorded at each epoch, which allowed progress to be checked for signs of overfitting.

A baseline classifier was used to provide a simple point of comparison for the CNN model. The baseline used in this project was a logistic regression classifier trained on flattened versions of the X-ray images. Unlike the CNN, this model does not learn spatial patterns or local features, instead it treats each pixel as an independent numerical value. As a result, it is able to provide a useful lower-bound performance level to show how much the CNN benefits from feature extraction through convolutional layers.

The purpose of including this baseline is to demonstrate whether the CNN actually improves classification performance compared to a simpler model. If the CNN performs significantly better, it supports the idea that learning visual patterns from the images leads to better results compared to relying solely on pixel values.

To evaluate the performance of the CNN model and compare it with the baseline classifier, accuracy, precision, recall and the F1-score were calculated on the test set. The CNN achieved the following results:

Metric	Value
Accuracy	0.737
Precision	0.710
Recall	0.979
F1-Score	0.823

Table 1. Performance metrics of the CNN model on the test set.

	Predicted: Normal	Predicted: Pneumonia
Actual: Normal	78	156
Actual: Pneumonia	8	382

Table 2. Confusion matrix for the CNN model on the test set.

The results from the CNN model show that it was able to learn useful patterns from the X-ray images and performed reasonably well on the test set. The strongest metric was Recall, which achieved 0.979 indicating that the model correctly identified almost all pneumonia cases. This is important in a medical context, as failing to detect pneumonia can delay treatment. The high recall is also consistent with the confusion matrix which shows only eight missed pneumonia cases.

The model's prediction was lower than the recall at 0.710 meaning that it produced a higher number of false positives. This is reflected in the confusion matrix where 156 normal images were incorrectly identified as having pneumonia. This behaviour suggests the model

has bias towards predicting pneumonia, which may be a result of a class imbalance or overfitting to possible pneumonia features during the training phase.

The training and validation accuracy curves also provide insight into the model's behaviour. Training accuracy increased steadily across all five epochs, while validation accuracy fluctuated, the highest being in epoch 4 followed by epoch 2. This pattern indicates that the model began to overfit the training data, likely due to the limited dataset size and the simplicity of the augmentation strategy.

When compared to the baseline logistic regression model, the CNN shows a clear advantage, as the regression model operates solely on raw pixel values and cannot capture spatial structure within images. While the CNN model used convolutional layers to extract hierarchical features, explaining to the better performance in recall and F1-score. This confirms that learning spatial patterns provides a meaningful improvement in pneumonia detection.

Overall, the results show that the CNN model is effective at identifying cases but tends to over-predict. While this reduces the risk of a missed diagnosis, it also increases the number of false positives.

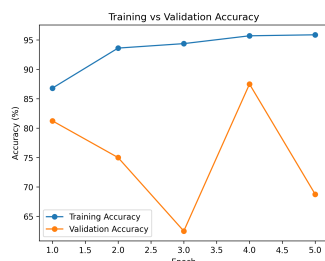


Figure 1. Training and validation accuracy across five epochs.

4 Discussion

The results demonstrate that the CNN was able to learn meaningful visual patterns from the chest X-rays and achieved a reasonable performance on the test set. The model produced a high recall, correctly identifying almost all pneumonia cases. This behaviour is desirable in a medical context[1], as missing a pneumonia diagnosis can lead to delayed treatment.

The biggest limitation of the current model is that it produces a large number of false positives. 156 images that were labelled as normal were misclassified as having pneumonia, indicating that the model shows bias towards predicting pneumonia. A likely explanation is class imbalance within the dataset[3] where pneumonia cases outnumbered normal cases.

Another reason for this bias could be overfitting, which is visible in the training and validation accuracy curve 1. Training accuracy increased consistently across all epochs, while the validation accuracy varied heavily and declined by the third epoch. This pattern suggests that the model began to memorise the training images rather than learning general features that transfer to unseen data.

The comparison with the baseline logistic regression model highlights the advantage of using convolutional layers for this task. Logistic regression treats all pixels independently and is unable to capture spatial relationships within the lungs.[7] While the CNN extracts these hierarchical features at different levels of complexity, the baseline classifier cannot, giving it the clear advantage in achieving higher recall and overall stronger performance.

The CNN shows clear potential for assisting with pneumonia detection, but improvements are needed to reduce the false-positive rate and to improve generalisation.

5 Conclusion and future work

This project demonstrated that a simple convolutional neural network can be used to classify chest X-ray images as either normal or pneumonia. The model achieved a strong recall and performed better than the baseline classifier. However, as it still produces a high number of false positives, the model is not yet reliable enough for real clinical use.

Future work would focus on improving the model's ability to generalise. The most promising way of doing this would be by expanding the dataset or by applying stronger data augmentation techniques, allowing the network to learn more robust features and reduce overfitting. Architectures such as ResNet and DenseNet are also examples of well-established models and could be explored to improve accuracy and stability beyond what is possible with a simplified CNN.[8] [5]

ACKNOWLEDGEMENTS

REFERENCES

- [1] Paulius Bundza and Justas Trinkūnas, 'Detection of pneumonia from chest x-rays using convolutional neural networks', *Mokslas-Lietuvos ateitis/Science-Future of Lithuania*, **17**, (2025).
- [2] GeeksforGeeks. Convolution and cross-correlation in cnn. <https://www.geeksforgeeks.org/machine-learning/convolution-and-cross-correlation-in-cnn/>, 2022. Accessed: 2025-02-10.
- [3] Hammad Iqbal, Arshad Khan, Narayan Nepal, Faheem Khan, and Yeon-Kug Moon, 'Deep learning approaches for chest radiograph interpretation: A systematic review', *Electronics*, **13**(23), 4688, (2024).
- [4] Paul Timothy Mooney. Chest x-ray pneumonia dataset. <https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>, 2018. Accessed: 2025-02-10.
- [5] Jalal Rabbah, Mohammed Ridouani, and Larbi Hassouni, 'Improving pneumonia diagnosis with high-accuracy cnn-based chest x-ray image classification and integrated gradient', *Biomedical Signal Processing and Control*, **101**, 107239, (2025).
- [6] Adrian Rosebrock. Convolution and cross-correlation in neural networks. <https://pyimagesearch.com/2021/05/14/convolution-and-cross-correlation-in-neural-networks/>, 2021. Accessed: 2025-02-10.
- [7] DR Sarvamangala and Raghavendra V Kulkarni, 'Convolutional neural networks in medical image understanding: a survey', *Evolutionary intelligence*, **15**(1), 1–22, (2022).
- [8] Houmem Slimi, Ala Balti, Sabeur Abid, and Mounir Sayadi, 'Trustworthy pneumonia detection in chest x-ray imaging through attention-guided deep learning', *Scientific Reports*, **15**(1), 40029, (2025).
- [9] PyTorch Team. Training a classifier. https://docs.pytorch.org/tutorials/beginner/blitz/cifar10_tutorial.html, 2024. Accessed: 2025-02-10.